STATE BOARD OF TECHNICAL EDUCATION KERALA

DIRECTORATE OF TECHNICAL EDUCATION

GOVERNMENT OF KERALA

Diploma in Engineering

Semester 1

STUDENT RECORD

1008 Foundational IT skills

In line with Diploma Curriculum Revision 2026



Prepared by:

STATE INSTITUTE OF TECHNICAL TEACHERS TRAINING AND RESEARCH, KALAMASSERY

STATE BOARD OF TECHNICAL EDUCATION KERALA

DIRECTORATE OF TECHNICAL EDUCATION

GOVERNMENT OF KERALA

Diploma in Computer Engineering

Semester 1

LABORATORY MANUAL

1008 - Foundational IT Skills

For Diploma Curriculum Revision 2026

(Student Copy)

Prepared by:
STATE INSTITUTE OF TECHNICAL TEACHERS TRAINING AND RESEARCH, KALAMASSERY

PREFACE

This laboratory manual is prepared for first-semester diploma students of all engineering branches as part of the Foundational IT Skills course under the Diploma Curriculum Revision 2026. The manual aims to build practical competency in essential IT areas including computer hardware, operating systems, office productivity tools, cloud collaboration, data analysis, and AI-assisted design — giving every student a strong digital foundation relevant to their programme and career.

Each experiment is aligned with the Course Outcomes (COs) and Program Outcomes (POs) of the curriculum, following a competency-based, student-centered approach. Students are expected to observe carefully, analyse results, and document their work in a structured manner.Google Classroom is the official platform for this course, used for assignment submission, resource sharing, and all instructor communications. Students must enroll using their institution email ID and remain active throughout the semester. This combined approach of maintaining handwritten records and submitting digital outputs prepares students for both traditional technical documentation and modern professional workflows.

PROGRAM OUTCOMES

POs	Statements
PO1	Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices
PO2	Identify and analyze well-defined engineering problems and reach substantiated conclusions using codified methods of analysis specific to their field of activity (DK1 to DK4).
PO3	Design solutions for well-defined technical problems and assist in the design of systems, components, or processes to meet specified needs, with appropriate consideration for public health and safety, as well as cultural, societal, and environmental factors (DK5).
PO4	Conduct investigations of well-defined problems by locating and search relevant codes and catalogues, and by conducting standard tests and measurements (DK8).
PO5	Apply appropriate techniques, resources, and modern computing, engineering and IT tools to well-defined engineering problems, while being aware of their limitations (DK2 and DK6).
PO6	When solving well-defined engineering problems, evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (DK1, DK5, and DK7).
PO7	Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).
PO8	Function effectively as an individual, and as a member or leader in diverse and inclusive teams, in multidisciplinary face-to-face/remote/distributed settings (DK9).
PO9	Communicate effectively and inclusively on well-defined engineering activities with the engineering community and society at large by comprehending the work of others, documenting one's own work, and giving and receiving clear instructions.
PO10	Demonstrate awareness of engineering management principles as a member or leader of a technical team and apply them to manage projects in multidisciplinary environments.
PO11	Recognize the need for, and have the ability for independent updating in the face of specialized technical knowledge (DK8).

KNOWLEDGE AND ATTITUDE PROFILE

DKs	Statements
DK1	A descriptive, formula-based understanding of the natural sciences applicable in a sub-discipline and awareness of directly relevant social sciences.
DK2	Procedural mathematics, numerical analysis, statistics applicable in a sub discipline.
DK3	A coherent procedural formulation of engineering fundamentals required in an accepted sub-discipline.
DK4	Engineering specialist knowledge that provides the body of knowledge for an accepted sub-discipline.
DK5	Knowledge that supports engineering design and operations based on the techniques and procedures of a practice area.
DK6	Codified practical engineering knowledge in recognized practice area.
DK7	Knowledge of issues and approaches in engineering technician practice, such as public safety and sustainable development.
DK8	Engagement with the current technological literature of the practice area.
DK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

COURSE DETAILS

ITEM	DETAILS	
Programme	Computer Engineering	
Semester	1	
Course Name	Foundational IT Skills	
Course Code	1008	
Course Type	Practicum / Laboratory	
Credits	1	
Hours/Week	2	
Total Hours	30	
Curriculum	Revision 2026	
Prepared By		
Institution		

COURSE OUTCOME MATRIX

CO	Course Outcome	Blooms Taxonomy Level
CO1	Identify the basic functions and features of Computers, Operating Systems, and Internet applications.	L2
CO2	Use Stand-alone and Cloud-based office tools to prepare documents, spreadsheets, and presentations.	L3
CO3	Apply spreadsheet functions and Generative AI tools to construct interactive dashboards, automated reports, and professional visual presentations for interpreting complex datasets.	L3
CO4	Apply generative AI, digital design platforms for guided learning and visual communication.	L3

COURSE ARTICULATION MATRIX

Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3										
CO2	3					3	3				
CO3	3	3		3			3				
CO4	3	3	3	3			3				
Total	12	6	3	6		3	9				
Correlation level	3	3	3	3		3	3				

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching Scheme/ Week				Self-learning (SS)/ Week	Total Hours/ Semester	Credits C	Examination Scheme						
L	T	P	R				Theory			Practical			Total
							CIE	ESE	Total	CIE	ESE	Total	
0	0	2	0	1	30	1	0	0	0	60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit)
SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

GENERAL INSTRUCTIONS

1. Rough record and Fair record/ Printed copy of report uploaded in Google classroom are needed to record the experiments conducted in the laboratory.
2. Rough records are needed to be certified immediately on completion of the experiment.
3. Fair records/ Report upload are due at the beginning of the next lab period.
4. Fair records/Printed copy of report uploaded in Google classroom must be submitted as neat, legible, and complete.

INSTRUCTIONS FOR WRITING THE FAIR RECORD/REPORT

In the fair record, the index page should be filled properly by writing the corresponding experiment number, experiment name, date on which it was done and the page number. On the **right-side** page of the record, the following has to be written:

1. **Title**: The title of the experiment should be written on the page in capital letters.
2. In the left top margin, **experiment number** and **date** should be written.
3. **Practical significance/Aim**: The purpose of the experiment should be written clearly.
4. **Relevant course outcome**: The outcome to be attained at the end of the experiment has to be mentioned clearly.
5. **Theory**: Simple working of the circuit/experimental set up/algorithm should be written.
6. **Procedure**: Steps for doing the experiment and recording the readings should be briefly described.
7. **Results**: The results of the experiment must be summarized in writing and should be fulfilling the aim.

On the **left-side** page of the record following has to be recorded:

1. **Output:** Relevant outputs shall be presented in written format and/or supported with screenshots wherever applicable.

GENERAL GUIDELINES TO TEACHERS

The experiments are developed such that the major practical learning outcomes are achieved. Each experiment is designed to achieve the course outcomes mentioned in the curriculum.

1. Teacher should provide the guideline with demonstration of experiment to the students with all features.
2. Teacher shall explain prior concepts to the students before the starting of each practical session.
3. Involve students in performance of each experiment.
4. Teacher is expected to share the skills and competencies to be developed in the students.
5. Teacher should ensure that the respective skills and competencies are developed in the students after the completion of the practical exercise.
6. Teacher should give opportunity to students for hands on experience after the demonstration.
7. Teacher may provide additional knowledge and skills to the students even though not covered in the manual but are expected from the students by the industry.
8. Teacher may suggest additional related literature of technical papers/reference books.
9. Finally give practical assignments and assess the performance of the students based on task assigned to check whether it is as per the instructions.
10. Teacher is expected to refer complete curriculum document and follow guidelines for implementation.
11. For difficult practical, if required, teacher could provide demonstration of the practical emphasizing of skills which the student should achieve.
12. Assess the skill achievement of the students and COs after each unit.
13. The assessment should be performed on regular basis.

GENERAL GUIDELINES TO STUDENTS

1. Listen carefully to all the information regarding curriculum, its course outcomes and major learning outcomes, equipment and instruments in the laboratory and the method of assessment.
2. Read the writeup of each experiment to be performed, a day advance.
3. Be **punctual** for your laboratory sessions.
4. It is mandatory to come to lab in a formal dress and wear your ID cards.
5. Mobile phones should be switched off in the lab. Keep bags in the bag rack.
6. Sign in the log register as soon as you enter the lab and strictly observe your lab timings.
7. Strictly follow the written and verbal instructions given by the teacher / Lab Instructor. If you do not understand the instructions, the handouts and the procedures, ask the instructor or teacher.
8. **Never work alone!** You should be accompanied by your laboratory partner and / or the instructors / teaching assistants all the time.
9. Organize the work in group and record all the observations.
10. Understand the practical implications of the experiments.
11. Students should develop maintenance and troubleshooting skills.
12. Keep the labs clean at all times; no food and drinks allowed inside the lab.
13. Intentional misconduct will lead to expulsion from the lab.
14. After completion of experiment, return the bread board, trainer kits, wires, CRO probes and other components to lab staff. Do not take any item from the lab without permission.
15. Observation book and lab record should be carried to each lab. Readings of current lab experiment are to be entered in observation book and previous lab experiment should be written in lab record book. Both the books should be corrected by the faculty in each lab.

SAFETY INSTRUCTIONS

1. Do not wear loose-fitting clothing or jewelry in the lab. Rings and necklaces are excellent conductors of electricity.
2. Sandals or open-toed shoes are NOT allowed.
3. Do not handle any equipment without reading the safety instructions. Read the handout and procedures in the lab manual before starting the experiments.
4. Setup the circuit and perform a careful circuit check before applying power. Do not make circuit changes or perform any wiring when power is on.
5. Avoid contact with energized electrical circuits.
6. Do not insert connectors forcefully into the sockets.
7. **NEVER** try to experiment with the power from the wall plug.
8. Immediately report dangerous or exceptional conditions to the Lab instructor / teacher: Equipment that is not working as expected, wires or connectors are broken, the equipment that smells or "smokes". If you are not sure what the problem is or what's going on, switch off the Emergency shutdown.
9. Never use damaged instruments, wires or connectors. Hand over these parts to the Lab instructor/Teacher.
10. Be sure of location of fire extinguishers and first aid kits in the laboratory.

ASSESSMENT METHODOLOGY

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			2 hours	2 hours		2.5 hours
Exam mark		50	40	40	50	40
Converted to			15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Continuous Internal Evaluation

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
TOTAL			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
TOTAL		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
TOTAL		50

CONTENT PAGE

Sl. No.	Title of the practical	Mapped CO	Page No.	Executed date	Remarks
1	Hardware & Software Exploration	CO1	1		
2	File and Folder Management in a Linux GUI	CO1	7		
3	Web & Connectivity	CO1	12		
4	Professional Documentation	CO2	18		
5	Spreadsheet Basics	CO2	23		
6	Presentation Design	CO2	28		
7	Cloud Collaboration	CO2	32		
8	Data Analysis	CO3	37		
9	Data Visualization	CO3	41		
10	AI Slide Generation	CO3	46		
11	Generative Pre-trained Transformer (GPT) Guided Learning	CO4	50		
12	Graphic Design	CO4	54		

Experiment no.: 1	Date:
HARDWARE & SOFTWARE EXPLORATION	

PRACTICAL SIGNIFICANCE

To identify the major hardware components of a computer system and familiarize students with system software and application software available on a personal computer.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices.

RELEVANT COURSE OUTCOMES(COS)

CO1 : Identify the basic functions and features of Computers, Operating Systems, and Internet applications.

THEORY

Write the minimum theory required for performing the experiment.

Hardware

Hardware refers to the physical components of a computer system that can be seen and touched.

Major hardware components include:

*1. **CPU** (Central Processing Unit)*

** CPU is called the "brain of the computer."*

** It processes instructions and performs calculations.*

** It controls all operations of the computer system.*

** Main components: Arithmetic Logic Unit (ALU), Control Unit (CU), Registers*

Functions:

** Executes instructions*

** Processes data*

** Controls system operations*

2. RAM (Random Access Memory)

** RAM is temporary memory used by the computer while executing programs.*

** Data stored in RAM is lost when power is switched off.*

Functions:

** Stores currently running programs*

** Stores temporary data*

Characteristics:

** Volatile memory*

** Faster than secondary storage*

3. HDD (Hard Disk Drive)

** HDD is a secondary storage device.*

** Stores operating systems, files, software, and user data permanently.*

Characteristics:

** Non-volatile storage*

** Mechanical storage device*

4. SSD (Solid State Drive)

** SSD is a storage device using flash memory.*

** It has no moving parts.*

Advantages over HDD:

** Faster access speed*

** Less power consumption*

** More durable*

5. Peripheral Ports

Peripheral ports are interfaces used to connect external devices to the computer.

Common ports:

Port	***Purpose***
USB Port	*Connect keyboard, mouse, pen drive, printer*
HDMI Port	*Connect monitor/projector*
VGA Port	*Connect display devices*
Ethernet Port	*Network connection*
Audio Port	*Connect speakers/headphones*
Power Port	*Supply power*
Type-C Port	*Data transfer and charging*

Software

Software is a collection of programs and instructions that directs the computer to perform specific tasks.

Software is classified into:

1. System Software

System software controls the operation of the computer and manages hardware resources.

Examples:

* *Operating System*

* *Device Drivers*

* *Language Translators*

* *Utility Software*

Examples available on PC:

* *Windows*

* *Linux*

* *Antivirus software*

* *Disk Cleanup utility*

* *Device drivers*

Functions:

* *Manages hardware*

** Provides user interface*

** Controls system resources*

2. Application Software

Application software performs specific user tasks.

Examples:

Software	***Purpose***
MS Word	*Document creation*
MS Excel	*Spreadsheet operations*
MS PowerPoint	*Presentation preparation*
Google Chrome	*Web browsing*
VLC Media Player	*Playing media files*
Paint	*Image editing and drawing.*
Calculator	*Calculations*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Monitor	15 inch or above	1
3	Keyboard	Standard USB keyboard	1
4	Mouse	Standard USB mouse	1
5	RAM Module	Any available module for demonstration	1
6	HDD / SSD	Any available drive for demonstration	1
7	Internet Connection	Broadband / Wi-Fi	1 per lab
8	Ubuntu Linux OS	Version 22.04 LTS or above	Installed
9	Web Browser	Mozilla Firefox / Google Chrome	Installed

PROCEDURE

❖ *Hardware Identification*

1. Observe the computer system carefully.

2. Identify CPU.

3. Identify RAM.

4. Identify HDD/SSD.

5. Observe peripheral ports available on the computer.

6. Note down the purpose of each port.

❖ *Software Identification*

1. Switch on the computer.

2. Open installed applications.

3. Identify system software.

4. Identify application software.

5. Record observations.

RESULTS

Thus the major hardware components and software components of a personal computer were identified and studied successfully.

PRACTICAL RELATED QUESTIONS

1. What is CPU?

2. Expand RAM.

3. What is the difference between RAM and ROM?

4. What is HDD?

5. What is SSD?

6. Name any two peripheral ports.

7. What is system software?

8. What is application software?

9. Give examples of system software.

10. Give examples of application software.

SELF- LEARNING COMPONENTS

1. Draw or Take a picture of the ports of your Lab PC and label the ports of your PC.

2. Compare the RAM, Processor, and Storage type (SSD vs. HDD) of your Tablet PC or any other smart devices versus your lab PC.Prepare a comparison chart of the configurations.

3. List any 5 system software and application software in your lab PC.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://www.geeksforgeeks.org/computer-science-fundamentals/computer-hardware/	GeeksforGeeks	GeeksforGeeks

Experiment no.: 2	Date:

FILE AND FOLDER MANAGEMENT IN LINUX GUI

PRACTICAL SIGNIFICANCE

To create and manage files and folders in Linux using the Graphical User Interface (GUI).

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices.

RELEVANT COURSE OUTCOMES(COS)

CO1 : Identify the basic functions and features of Computers, Operating Systems, and Internet applications.

THEORY

Write the minimum theory required for performing the experiment.

Linux Operating System

Linux is an open-source operating system that manages computer hardware and software resources. It supports both command-line and graphical user interfaces.

Features of Linux:

Open-source operating system

Multi-user operating system

Multi-tasking capability

Secure and reliable

Supports GUI and command line interface

Graphical User Interface (GUI)

GUI allows users to interact with the computer using windows, icons, menus, and a mouse pointer.

Advantages of GUI:

Easy to use

User-friendly

Does not require memorizing commands

Supports drag-and-drop operations

File

A file is a collection of information stored on a storage device.

Examples:

Text files,Images,Audio files, Video files, Documents

Folder (Directory)

A folder or directory is used to organize and store files and other folders.

Functions of folders:

Organize files

Reduce confusion

Improve file management

Common File Management Operations

Create : Creating a new file or folder.

Rename : Changing the name of an existing file or folder.

Copy: Creating a duplicate of a file or folder.

Move: Changing the location of a file or folder.

Delete: Removing unwanted files or folders.

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Ubuntu Linux OS	Version 22.04 LTS or above	Installed
3	Files Application (File Manager)	GNOME Files / Nautilus (default in Ubuntu)	Installed
4	Text Editor	Gedit / Mousepad	Installed

PROCEDURE

Creating Folders

1. *Open File Manager.*
2. *Navigate to Home folder.*

3. *Right-click in empty space.*
4. *Select New Folder.*
5. *Create folder named ComputerLab.*

Creating Files

1. *Open ComputerLab folder.*
2. *Create files:*
    - *Student1.txt*
    - *Student2.txt*
    - *Student3.txt*

Rename Operation

1. *Rename Student1.txt as StudentData.txt.*

Copy Operation

1. *Copy Student2.txt into another folder.*

Move Operation

1. *Move Student3.txt to Documents folder.*

Delete Operation

1. *Delete unwanted files.*

LINUX COMMANDS RELATED TO FILE AND FOLDER MANAGEMENT (Optional Exercise For Computer allied branches)

Although file and folder operations can be performed using the Graphical User Interface (GUI), Linux also provides command-line utilities for performing the same tasks efficiently. Understanding these commands helps students gain familiarity with Linux system administration and file management.

1. pwd – Print Working Directory

Purpose: Displays the current directory path.

2. ls – List Directory Contents

Purpose: Displays files and folders in the current directory.

Useful Options: ls -l, ls -a

3. mkdir – Create Directory

Purpose: Creates a new folder (directory).

4. touch – Create Empty File

Purpose:Creates a new empty file.

5. cd – Change Directory

Purpose:Changes the current working directory.

6. cp – Copy Files and Directories

Purpose:Creates a duplicate copy of a file or folder.

7. mv – Move or Rename Files

Purpose:Moves files/folders or renames them.

8. rm – Remove Files

Purpose:Deletes files.

9. rmdir – Remove Empty Directory

Purpose: Deletes an empty folder.

10. rm -r – Remove Directory and Contents

Purpose:Deletes a folder along with all files and subfolders.

11. cat – Display File Contents

Purpose: Displays the contents of a text file.

Displays the file contents on the terminal.

12. clear – Clear Terminal Screen

Purpose:Clears all text displayed in the terminal window.

Practical Note:

The instructor of Computer allied branches may first demonstrate file and folder management using the GUI File Manager and then show the equivalent Linux terminal commands so that students understand both graphical and command-line approaches to file management. This helps bridge GUI-based learning with Linux system administration concepts.

RESULTS

File and folder management operations such as creating, renaming, copying, moving, and deleting files and folders were successfully performed using the Linux Graphical User

Interface (GUI). The student gained familiarity with the Linux desktop environment and file management tools. A report containing the steps performed was prepared and successfully uploaded through Google Classroom.

PRACTICAL RELATED QUESTIONS

1. *What is Linux?*
2. *Expand GUI.*
3. *What is a file?*
4. *What is a folder?*
5. *What is File Manager?*
6. *How do you create a folder?*
7. *What is the difference between copy and move?*
8. *What is rename operation?*
9. *What is the purpose of Trash?*
10. *Name two Linux desktop environments.*

SELF- LEARNING COMPONENTS

1. Use the GUI to create a directory tree for your semester subjects, using subject codes as folder names and creating separate sub-folders to organize notes for each module.

2. Create folders and files for the given scenario

You are preparing for a weekend trip. Your "home base" is your main user folder.Create a new folder called Backpack.Go inside your Backpack folder and create two folders: Food and Tools.Inside the Tools folder, create a file called flashlight.txt and a file called map.txt. You realize the map.txt shouldn't be in Tools. Move it into the Food folder instead.You want a second flashlight just in case. Copy the flashlight.txt file and name the copy spare_light.txt. You decide you don't need the Food folder at all. Delete the entire Food folder and everything inside it.List everything left in your Backpack to see what you have.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://ubuntu.com/tutorials/command-line-for-beginners	Canonical Ltd.	Ubuntu Official
2	https://www.geeksforgeeks.org/linux-file-hierarchy-structure/	GeeksforGeeks	GeeksforGeeks

Experiment no.: 3	Date:
WEB & CONNECTIVITY	

PRACTICAL SIGNIFICANCE

To practice effective web searching using advanced search operators, understand email etiquette, and familiarize students with the use of Google Classroom.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices.

RELEVANT COURSE OUTCOMES(COS)

CO1 : Identify the basic functions and features of Computers, Operating Systems, and Internet applications.

THEORY

Web connectivity refers to the ability of a computer system to connect to the internet and access online resources and services.

Requirements for web connectivity:

** Computer or mobile device*

** Internet connection*

** Web browser*

** User account (for online services)*

Examples of web browsers:

** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari*

Advanced Search Operators

Search operators are special commands used with search engines to obtain more accurate and filtered results.

Common search operators:

Operator	*Purpose*	*Example*
" "	*Search exact phrase*	*"computer networks"*
site:	*Search within a specific website*	*site:edu operating system*
filetype:	*Search for a particular file type*	*filetype:pdf Linux tutorial*
OR	*Search either keyword*	*Python OR Java*
-	*Exclude a word*	*Java -coffee*
***	*Wildcard operator*	*"largest * in India"*

Advantages:

** Reduces irrelevant search results*

** Saves time*

** Provides accurate information*

** Helps locate specific documents*

Email Etiquette

Email etiquette refers to the professional and proper behavior followed while sending and receiving emails.

Guidelines for proper email writing:

1. Use a clear subject line

Example:

Subject: Submission of Lab Record

2. Begin with proper greeting

Examples:

* *Dear Sir/Madam,*

* *Respected Sir,*

* *Hello Team,*

3. Write concise and clear messages

* *Use simple language*

* *Keep the content relevant*

* *Avoid unnecessary information*

4. Maintain professional language

Avoid:

* *Informal words*

* *Slang language*

* *Excessive punctuation*

5. End politely

Examples:

* *Thank you*

* *Regards*

* *Yours faithfully*

6. Check before sending

Verify:

* *Recipient address*

* *Attachments*

* *Grammar*

* *Spelling mistakes*

Google Classroom

Google Classroom is an online learning platform used to manage classes and assignments digitally.

Features:

** Join classes using class code*

** View assignments*

** Submit assignments*

** Download study materials*

** Communicate with instructors*

** Receive notifications*

Advantages:

** Easy content sharing*

** Paperless submission*

** Better communication*

** Organized learning environment*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab
3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account	Valid institution or personal email ID	1 per student
5	Google Classroom	Accessed via browser	1 per student

PROCEDURE

List the one-by-one procedure to be followed for completing the experiment.

Activity 1: Advanced Search Operators

1. Open a web browser.

2. Open a search engine.

3. Practice the following searches:

** "Computer Networks"*

** site:edu computer engineering*

** filetype:pdf Linux notes*

** Python OR Java*

** Java -coffee*

4. Observe search results.

Activity 2: Email Etiquette

1. Open email application.

2. Compose a new email.

3. Enter recipient email address.

4. Enter subject.

5. Write a proper message.

6. Attach a file if required.

7. Verify details and send.

Activity 3: Google Classroom

1. Open Google Classroom.

2. Login using Google account.

3. Join a class using class code.

4. Access learning materials.

5. Submit a sample assignment.

RESULTS

Thus advanced search operators, email etiquette practices, and Google Classroom functionalities were studied and practiced successfully.

PRACTICAL RELATED QUESTIONS

1. What is a search operator?

2. What does site: operator do?

3. What is filetype: operator?

4. Why is email etiquette important?

5. What is Google Classroom?

6. Name two web browsers.

SELF- LEARNING COMPONENTS

1. Use filetype:pdf "Computer Engineering" site:edu on Google. Record the URL and File Size (KB) of the first PDF result found.

2. Enroll to google classroom or any other LMS of Foundational IT Skills and customize profile.

3. Create an email id, draft a formal mail and send mail to your friends with netiquitte.(scenario- You are the Event Coordinator for your college. You need to send a formal announcement to the Staff coordinator, 10 student participants regarding a change in the workshop schedule. You must also keep the HoD informed of this update, but you must ensure the students' private email addresses are not visible to one another.)

REFERENCES

Sl. No.	Text Book/ Website
1	https://ahrefs.com/blog/google-advanced-search-operators/

Experiment no.: 4	Date:

PROFESSIONAL DOCUMENTATION

PRACTICAL SIGNIFICANCE

To create a professional document using a word processing application with proper formatting, styles, and an automatically generated table of contents.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices.

PO6: When solving well-defined engineering problems, evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (DK1, DK5, and DK7).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO2: Use Stand-alone and Cloud-based office tools to prepare documents, spreadsheets, and presentations.

THEORY

Professional Documentation

Professional documentation refers to preparing documents in a structured and organized manner for communication and presentation purposes.

Examples:

** Project reports, Seminar reports, Lab records, Technical documentation, Business reports*

Characteristics of a professional document:

** Proper title and headings*

** Consistent formatting*

** Page numbering*

** Table of contents*

** Proper font and alignment*

** Well-organized sections*

Word Processing Software

Word processing software allows users to create, edit, format, and print documents.

Examples:

** Microsoft Word*

** LibreOffice Writer*

** Google Docs*

Functions:

** Text editing*

** Formatting*

** Inserting images and tables*

** Spell checking*

** Document organization*

Styles

Styles are predefined formatting settings used to maintain consistency throughout a document.

Common styles:

Style	*Purpose*
Title	*Main document title*
Heading 1	*Main headings*
Heading 2	*Subheadings*

Normal	*Regular text*
Subtitle	*Secondary titles*

Advantages of using styles:

* *Uniform appearance*

* *Easy formatting*

* *Faster editing*

* *Automatic table of contents generation*

Table of Contents (TOC)

A table of contents is a list of document headings and corresponding page numbers.

Advantages:

* *Helps locate sections quickly*

* *Improves document organization*

* *Provides professional appearance*

* *Updates automatically when document content changes*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	LibreOffice Writer	Version 7.0 or above	Installed
3	Internet Connection	Broadband / Wi-Fi	1 per lab
4	Google Account	Valid institution or personal email ID	1 per student
5	Google Classroom	Accessed via browser	1 per student

PROCEDURE

Activity: Creating a Formal Report

1. Open the word processing application.

2. Create a new document.

3. Enter report title.

4. Add headings and content.

5. Apply styles:

** Title style for title*

** Heading 1 for main headings*

** Heading 2 for subheadings*

** Normal style for body text*

6. Insert page numbers.

7. Generate an automatic table of contents.

8. Save the document.

Sample Report Structure

Title:Computer Fundamentals Report

1. Introduction

2. Hardware Components

3. Software Components

4. Advantages of Computers

5. Applications of Computers

6. Conclusion

RESULTS

Thus a formal report was created successfully using styles and an automatically generated table of contents.

PRACTICAL RELATED QUESTIONS

1. What is professional documentation?

2. What is a style?

3. Why are styles used?

4. What is TOC?

5. Which style is used for main headings?

6. Name two word processing software applications.

7. Why is formatting important?

8. How can a table of contents be generated?

9. What is the use of page numbering?

10. State one advantage of professional documentation.

SELF- LEARNING COMPONENTS

1. Create a standard 1-page "Assignment Cover Sheet" using Word Styles (Heading 1, Heading 2).

2. Create a professional Biodata for an internship application and export it into pdf (The Specification:

Page Layout: Margins: Set to "Normal" (1 inch or 2.54 cm on all sides).

Font: Use a professional sans-serif font (e.g., Arial or Calibri) for the body.

Use of Styles: Apply Heading 1 to main sections (Personal Details, Education, Skills).

Modify the Heading 1 style to be Bold and Underlined.

Data Organization (The Table): Use a 2-column Table to align your personal information (e.g., Date of Birth, Nationality, Languages) so the colons (:) line up perfectly.

Border: Remove the table borders so the text looks floating but perfectly aligned.

Visual Element:Insert a placeholder for a professional photo.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://help.libreoffice.org/latest/en-US/text/swriter/guide/main.html	The Document Foundation	LibreOffice Official Help

Experiment no.: 5	Date:

SPREADSHEET BASICS

PRACTICAL SIGNIFICANCE

To understand spreadsheet operations and use formulas such as SUM, AVERAGE (AVG), COUNT, COUNTIF, and IF with cell referencing.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices.

PO6: When solving well-defined engineering problems, evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (DK1, DK5, and DK7).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO2: Use Stand-alone and Cloud-based office tools to prepare documents, spreadsheets, and presentations.

THEORY

Spreadsheet

A spreadsheet is an electronic worksheet used for organizing, storing, and performing calculations on data.

Examples of spreadsheet software:

Microsoft Excel, LibreOffice Calc, Google Sheets

Applications of spreadsheets:

- *Mark list preparation*
- *Attendance records*
- *Payroll processing*
- *Budget preparation*
- *Data analysis*

Components of a Spreadsheet

Workbook

A workbook is a collection of worksheets.

Worksheet

A worksheet consists of rows and columns used for entering data.

Row

Horizontal arrangement of cells

Identified by numbers

Example:

1, 2, 3, ...

Column

Vertical arrangement of cells

Identified by letters

Example:

A, B, C, ...

Cell

Intersection of row and column

Stores data

Example:

A1, B2, C5

Cell Referencing

Cell referencing is used to refer to a cell address in formulas.

Types of cell references:

1. Relative Reference

Changes automatically when copied.

Example:

=A1+B1

2. Absolute Reference

Does not change when copied.

Uses $ symbol.

Example:

=$A$1+$B$1

3. Mixed Reference

Either row or column remains fixed.

Examples:

=$A1

=A$1

Spreadsheet Functions

SUM Function

Used to calculate total values.

Syntax:

=SUM(range)

Example:

=SUM(B2:B6)

AVERAGE Function (AVG)

Used to calculate average values.

Syntax:

=AVERAGE(range)

Example:

=AVERAGE(B2:B6)

COUNT Function

Counts the number of cells containing numeric values.

Syntax:

=COUNT(range)

Example:

=COUNT(B2:B6)

COUNTIF Function

Counts cells that satisfy a condition.

Syntax:

=COUNTIF(range,condition)

Example:

=COUNTIF(B2:B6,">50")

IF Function

Performs logical testing.

Syntax:

=IF(condition,value_if_true,value_if_false)

Example:

=IF(B2>=50,"Pass","Fail")

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	LibreOffice Calc/ Google sheets	LibreOffice Calc Version 7.0 or above	Installed
3	Google Classroom	Accessed via browser	1 per student

PROCEDURE

Activity: Using Spreadsheet Functions

1. *Open spreadsheet software.*
2. *Create a new worksheet.*
3. *Enter student details and marks.*
4. *Apply formulas:*
   - *SUM*
   - *AVERAGE*
   - *COUNT*
   - *COUNTIF*
   - *IF*
5. *Apply cell references.*
6. *Observe the results.*
7. *Save the worksheet.*

RESULTS

Spreadsheet formulas (SUM, AVERAGE, COUNT, COUNTIF, and IF) and cell referencing were studied and applied successfully and uploaded the spreadsheet in the Google classroom.

PRACTICAL RELATED QUESTIONS

1. *What is a spreadsheet?*
2. *What is a workbook?*
3. *What is a worksheet?*
4. *What is cell referencing?*
5. *What is the syntax of SUM function?*
6. *What is COUNT function?*
7. *What is COUNTIF used for?*
8. *What is an IF function?*
9. *Differentiate between relative and absolute references.*
10. *Name two spreadsheet applications.*

SELF- LEARNING COMPONENTS

1. Find and apply the COUNTIF formula in a spreadsheet to count how many students passed a test

2. In a cricket database, use the SUM and AVERAGE functions to calculate the total and mean values for the "Runs Scored" field in the range B2:B21. Next, apply the COUNTIF formula to the "Player Category" field in range C2:C21 to find the total number of players listed as "Batsman." In the "Selection Status" column, use an IF formula to check the "Runs Scored" in cell B2 and display "Selected" if the value is 50 or higher, otherwise display "Waitlist." Finally, use the COUNT function on the "Player Name" field in range A2:A21 to verify the total number of participants in the academy.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://help.libreoffice.org/latest/en-US/text/scalc/guide/main.html	The Document Foundation	LibreOffice Official Help
2	https://www.geeksforgeeks.org/basic-excel-formulas/	GeeksforGeeks	GeeksforGeeks

Experiment no.: 6	Date:

PRESENTATION DESIGN

PRACTICAL SIGNIFICANCE

To create a professional 5-slide pitch deck using LibreOffice Impress in Ubuntu by applying themes, animations, and slide transitions.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices.

PO6: When solving well-defined engineering problems, evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (DK1, DK5, and DK7).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO2: Use Stand-alone and Cloud-based office tools to prepare documents, spreadsheets, and presentations.

THEORY

Presentation Software

Presentation software is used to create visual presentations containing text, images, graphics, charts, and multimedia objects.

Examples:

- *LibreOffice Impress*
- *Microsoft PowerPoint*
- *Google Slides*

Pitch Deck

A pitch deck is a short presentation used to communicate ideas, projects, products, or solutions effectively.

A typical pitch deck contains:

- *Title*
- *Problem Statement*
- *Solution*
- *Features/Benefits*
- *Conclusion*

Animation

Animation is the visual effect applied to objects within a slide.

Examples:

- *Appear*
- *Fade In*
- *Fly In*
- *Zoom*

Advantages:

- *Makes presentations attractive*
- *Highlights important information*
- *Improves audience engagement*

Slide Transition

Slide transition is the visual effect displayed when moving from one slide to another.

Examples:

- *Fade*
- *Push*
- *Wipe*
- *Split*

Advantages:

- *Improves presentation flow*
- *Gives a professional appearance*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	LibreOffice Impress	Version 7.0 or above	Installed

PROCEDURE

Activity: Creating a 5-Slide Pitch Deck

1. *Open LibreOffice Impress.*
2. *Create a new presentation.*
3. *Select a suitable theme.*
4. *Create five slides.*
5. *Add content to each slide.*
6. *Insert images if required.*
7. *Apply animations to objects.*
8. *Apply slide transitions.*
9. *Save the presentation.*
10. *Run the slideshow.*

RESULTS

A visual presentation containing text, images, graphics and multimedia objects was created successfully using presentation software and uploaded the pdf format of the same in the Google Classroom.

PRACTICAL RELATED QUESTIONS

1. *What is presentation software?*
2. *What is a pitch deck?*
3. *What is animation?*
4. *What is a transition?*
5. *Name two animation effects.*
6. *Name two transition effects.*
7. *Which application is used in Ubuntu for presentations?*
8. *What is the file extension of LibreOffice Impress?*
9. *Which key starts slideshow mode?*
10. *State one advantage of presentation software.*

SELF- LEARNING COMPONENTS

1. Edit a "Slide Master" to ensure your name and roll number appear automatically on header or footer of every slide of a deck.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://help.libreoffice.org/latest/en-US/text/simpress/guide/main.html	The Document Foundation	LibreOffice Official Help

Experiment no.: 7	Date:

CLOUD COLLABORATION

PRACTICAL SIGNIFICANCE

To understand cloud-based collaboration by using real-time editing features in Google Docs and Google Sheets and to explore version history management.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices.

PO6: When solving well-defined engineering problems, evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (DK1, DK5, and DK7).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO2: Use Stand-alone and Cloud-based office tools to prepare documents, spreadsheets, and presentations.

THEORY

Cloud Collaboration

Cloud collaboration is the process of allowing multiple users to access, edit, and share files or resources through internet-based platforms.

Advantages:

* *Enables real-time teamwork*

* *Automatic saving of work*

* *Access from any device*

* *Easy sharing of files*

* *Maintains version history*

Examples of cloud collaboration platforms:

** Google Docs*

** Google Sheets*

** Google Drive*

** Microsoft OneDrive*

** Dropbox*

Google Docs

Google Docs is a web-based document editor used to create and edit text documents online.

***Features**:*

** Real-time editing*

** Auto-save*

** Sharing and permissions*

** Comments and suggestions*

** Version history*

***Applications**:*

** Report writing*

** Documentation*

** Group assignments*

** Project work*

Google Sheets

Google Sheets is an online spreadsheet application used to create and analyze data.

***Features**:*

** Spreadsheet calculations*

** Real-time collaboration*

** Formula support*

** Chart creation*

** Version history*

***Applications**:*

** Mark lists*

** Attendance sheets*

** Budget preparation*

** Data analysis*

Version History

Version history stores previous changes made to a document.

***Advantages**:*

** Track changes made by users*

** Restore previous versions*

** Identify contributors*

** Prevent accidental data loss*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab
3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account	Valid institution or personal email ID	1 per student
5	Google Docs	Accessed via browser (docs.google.com)	1 per student

6	Google Sheets	Accessed via browser (sheets.google.com)	1 per student
7	Google Classroom	Accessed via browser	1 per student

PROCEDURE

Activity: Real-Time Editing and Version History

1. Open a web browser in Ubuntu.

2. Login to Google account.

3. Open Google Docs.

4. Create a new document.

5. Enter sample content.

6. Share the document with another user.

7. Perform collaborative editing.

8. Add comments and suggestions.

9. Open Google Sheets and create sample data.

10. View version history.

11. Restore an earlier version if required.

RESULTS

Thus real-time collaboration using Google Docs and Google Sheets along with version history features was studied and practiced successfully and uploaded the report in Google classroom.

PRACTICAL RELATED QUESTIONS

1. What is cloud collaboration?

2. What is Google Docs?

3. What is Google Sheets?

4. What is version history?

5. What is real-time editing?

SELF- LEARNING COMPONENTS

1. Imagine a team of four students is editing a single Google Sheet for a lab report. One student accidentally deletes a large data table and then closes the browser. Explain how the team can use the 'Version History' tool to see the names of people who edited the file and describe the steps to restore the table back to its original state.

2. In Google Sheets, if three people are working on a project, how can you tell which specific person deleted a value in Cell B5? Name the menu path you would use to find this information and explain how the 'color-coded' system helps identify the person.

3. Create a 1-page overleaf document that includes a title, a mathematical formula, and a list.(Optional Assignment)

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://support.google.com/docs/answer/2494822	Google	Google Docs Help Centre
2	https://support.google.com/docs/answer/190843	Google	Google Docs Help Centre

Experiment no.: 8	Date:

DATA ANALYSIS

PRACTICAL SIGNIFICANCE

To analyze a dataset using Pivot Tables and Slicers in Google Sheets for summarizing and filtering information.

RELEVANT PROGRAM OUTCOMES (POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices

PO2: Identify and analyze well-defined engineering problems and reach substantiated conclusions using codified methods of analysis specific to their field of activity (DK1 to DK4).

PO4: Conduct investigations of well-defined problems by locating and search relevant codes and catalogues, and by conducting standard tests and measurements (DK8).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO3: Apply spreadsheet functions and Generative AI tools to construct interactive dashboards, automated reports, and professional visual presentations for interpreting complex datasets.

THEORY

Data Analysis

Data analysis is the process of organizing and interpreting data to extract useful information.

Applications:

** Student performance analysis*

** Attendance analysis*

** Sales analysis*

** Survey reports*

** Budget planning*

Pivot Table

A Pivot Table is a spreadsheet feature used to summarize and rearrange large amounts of data dynamically.

Functions:

** Calculate totals*

** Calculate averages*

** Count records*

** Group data*

** Rearrange information*

Advantages:

** Faster analysis*

** Easy report generation*

** Better understanding of data*

Slicer

A Slicer is a visual filtering tool used with Pivot Tables.

Functions:

** Filters data interactively*

** Displays selected categories*

** Makes data analysis easier*

Advantages:

** User-friendly*

** Fast filtering*

** Improves visualization*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab
3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account	Valid institution or personal email ID	1 per student
5	Google Sheets	Accessed via browser (sheets.google.com)	1 per student
6	Google Classroom	Accessed via browser	1 per student

PROCEDURE

Activity: Data Analysis Using Pivot Tables and Slicers

1. Open Google Sheets.

2. Create a blank spreadsheet.

3. Enter sample data.

4. Select the dataset.

5. Create a Pivot Table.

6. Configure row, column, and value fields.

7. Add a Slicer.

8. Filter the data.

9. Analyze the results.

10. Save the spreadsheet.

RESULTS

The dataset was successfully analyzed using Pivot Tables and Slicers in Google Sheets to summarize, organize, and filter information. The observations and analysis results were

documented in a report containing the dataset, Pivot Table output, Slicer-based filtering results, interpretations, and necessary screenshots. The completed report was successfully uploaded and submitted through Google Classroom.

PRACTICAL RELATED QUESTIONS

1. What is a Pivot Table?

2. What is a Slicer?

3. Why is data analysis performed?

4. What is filtering?

5. What is the purpose of Pivot Tables?

6. Name one advantage of Slicers.

7. What is the use of row fields?

8. What is the use of value fields?

9. Which application was used for creating slicers and pivot table?

10. State one application of data analysis.

SELF- LEARNING COMPONENTS

1. Create a Pivot Table from a 50-row sales dataset. Drag "Category" to Rows and "Sales" to Values. Record the Grand Total value displayed.

2. You have a list of sales for 'Laptop', 'Mouse', and 'Keyboard'. Create a Pivot Table to show the total sales for each. Insert a Slicer for 'Region'. If you click 'South' on the Slicer, what happens to the 'Grand Total'? If you add a new row of data to your original list, name the button you must click to update the Pivot Table.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://support.google.com/docs/answer/1272900	Google	Google Sheets Help Centre
2	https://www.geeksforgeeks.org/google-sheets/how-to-create-and-use-pivot-tables-in-google-sheets/	GeeksforGeeks	GeeksforGeeks

Experiment no.: 9	Date:
DATA VISUALIZATION	

PRACTICAL SIGNIFICANCE

To create and analyze different types of charts in a spreadsheet for graphical representation and interpretation of data.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices

PO2: Identify and analyze well-defined engineering problems and reach substantiated conclusions using codified methods of analysis specific to their field of activity (DK1 to DK4).

PO4: Conduct investigations of well-defined problems by locating and search relevant codes and catalogues, and by conducting standard tests and measurements (DK8).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO3: Apply spreadsheet functions and Generative AI tools to construct interactive dashboards, automated reports, and professional visual presentations for interpreting complex datasets.

THEORY

Write the minimum theory required for performing the experiment.

Data Visualization

Data visualization is the graphical representation of data and information using charts, graphs, and diagrams to understand trends, patterns, and comparisons.

Advantages:

** Simplifies large amounts of data*

** Helps identify trends and patterns*

** Makes comparisons easier*

** Improves understanding*

** Supports decision-making*

Applications:

** Student performance analysis*

** Sales analysis*

** Attendance analysis*

** Survey reports*

** Financial reports*

Types of Charts

1. Column Chart

Displays data using vertical bars.

Applications:

** Comparing values among categories*

** Comparing student marks*

2. Bar Chart

Displays data using horizontal bars.

Applications:

** Ranking information*

** Comparing categories with longer names*

3. Pie Chart

Displays percentage distribution of values.

Applications:

** Showing proportions*

** Category contribution analysis*

4. Line Chart

Displays trends and changes over time.

Applications:

** Attendance trends*

** Monthly performance analysis*

** Sales reports*

5. Scatter Chart

Displays relationships between two variables.

Applications:

** Correlation studies*

** Data comparison*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab
3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account	Valid institution or personal email ID	1 per student
5	Google Sheets	Accessed via browser (sheets.google.com)	1 per student
6	Google Classroom	Accessed via browser	1 per student

PROCEDURE

Activity: Creating Different Types of Charts

1. Open Google Sheets.

2. Create a blank spreadsheet.

3. Enter sample data.

4. Select the dataset.

5. Open Insert → Chart.

6. Create:

** Column Chart*

** Bar Chart*

** Pie Chart*

** Line Chart*

** Scatter Chart*

7. Add titles and labels.

8. Analyze the generated charts.

9. Save the spreadsheet.

RESULTS

Thus different types of charts were created and analyzed successfully using a spreadsheet application.

PRACTICAL RELATED QUESTIONS

1. What is data visualization?

2. What is a chart?

3. What is the purpose of a Pie Chart?

4. Which chart is used for trends?

5. Which chart is suitable for comparison?

6. What is a legend?

7. What is the use of chart title?

8. What is a Scatter Chart?

9. Which application was used?

10. State one advantage of data visualization.

SELF- LEARNING COMPONENTS

1. Create a small summary table showing the count of students who 'Passed' (Score ≥ 40) versus 'Failed' (Score < 40) in the Series Test of Foundational IT Skills. Insert a 3D Pie Chart to represent these two groups as percentages of the whole class.

2. Create a Clustered Bar Chart (Horizontal) that displays the students' names on the Y-axis and their marks on the X-axis. After creating the chart, apply a Chart Style from the Design tab to change the background to a dark theme.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://support.google.com/docs/answer/63824	Google	Google Sheets Help Centre
2	https://www.geeksforgeeks.org/data-visualization/types-of-graphs-in-statistics/	GeeksforGeeks	GeeksforGeeks

Experiment no.: 10	Date:

AI SLIDE GENERATION

PRACTICAL SIGNIFICANCE

To generate professional multi-slide presentations using Generative AI platforms by providing structured text prompts.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices

PO2: Identify and analyze well-defined engineering problems and reach substantiated conclusions using codified methods of analysis specific to their field of activity (DK1 to DK4).

PO4: Conduct investigations of well-defined problems by locating and search relevant codes and catalogues, and by conducting standard tests and measurements (DK8).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO3: Apply spreadsheet functions and Generative AI tools to construct interactive dashboards, automated reports, and professional visual presentations for interpreting complex datasets.

THEORY

Generative AI

Generative AI is a type of Artificial Intelligence that creates new content such as text, images, presentations, and reports based on user input.

Applications:

* *Presentation generation*

* *Report generation*

* *Content creation*

* *Image generation*

** Educational support*

Advantages:

** Saves time*

** Improves productivity*

** Generates organized content*

** Supports creativity*

AI Slide Generation

AI slide generation refers to creating presentation slides automatically from user prompts.

Basic workflow:

1. Enter prompt

2. AI generates slide structure

3. AI creates content and design

4. User edits output

5. Presentation is exported

Prompt Engineering for Presentation Generation

A structured prompt should contain:

** Topic*

** Number of slides*

** Target audience*

** Main sections*

** Style of presentation*

Example Prompt:

"Create a 5-slide presentation on Cloud Computing for diploma students. Include introduction, applications, advantages, challenges, and conclusion using professional style."

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab
3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account	Valid institution or personal email ID	1 per student
5	Gamma AI	Accessed via browser (gamma.app)	1 per student
6	Google Classroom	Accessed via browser	1 per student

PROCEDURE

Activity: Generating Slides using Gamma AI

1. Open web browser in Ubuntu.

2. Open Gamma AI.

3. Sign in.

4. Create a new presentation.

5. Enter a structured prompt.

6. Generate outline.

7. Generate slides.

8. Review content.

9. Modify slides if required.

10. Export presentation.

RESULTS

A professional multi-slide presentation was successfully generated using a Generative AI platform by providing structured text prompts. The generated presentation was reviewed,

refined, and customized as required. A report containing the prompt used, screenshots of the AI generation process, screenshots of the final presentation was prepared and successfully uploaded through Google Classroom.

PRACTICAL RELATED QUESTIONS

1. What is Generative AI?

2. What is Gamma AI?

3. What is a prompt?

4. What is prompt engineering?

5. Why should prompts be specific?

6. Why should AI-generated content be reviewed?

7. Name one AI presentation platform.

8. What is the purpose of exporting presentations?

9. What are the advantages of AI slide generation?

10. State one application of Generative AI.

SELF- LEARNING COMPONENTS

1. Use Gamma.app to generate an 8-slide deck. Record the time taken (in seconds) for the AI to generate the full presentation.

2. Use Manus AI to research the current market price of three different laptop brands (Dell, HP, and Lenovo) with 16GB RAM. Ask the agent to summarize these prices into a formatted table and then create a 5-sentence summary recommending the best value for a student budget and make a presentation.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://help.gamma.app/en/	Gamma Technologies Inc.	Gamma Official Help
2	https://www.geeksforgeeks.org/artificial-intelligence/what-is-generative-ai/	GeeksforGeeks	GeeksforGeeks

Experiment no.: 11	Date:

GENERATIVE PRE-TRAINED TRANSFORMER (GPT) GUIDED LEARNING

PRACTICAL SIGNIFICANCE

To create and use a structured "Study Assistant" prompt in a Generative Pre-trained Transformer (GPT) system for learning technical concepts effectively.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices

PO2: Identify and analyze well-defined engineering problems and reach substantiated conclusions using codified methods of analysis specific to their field of activity (DK1 to DK4).

PO3: Design solutions for well-defined technical problems and assist in the design of systems, components, or processes to meet specified needs, with appropriate consideration for public health and safety, as well as cultural, societal, and environmental factors (DK5).

PO4: Conduct investigations of well-defined problems by locating and search relevant codes and catalogues, and by conducting standard tests and measurements (DK8).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO4: Apply generative AI, digital design platforms for guided learning and visual communication.

THEORY

Generative Pre-trained Transformer (GPT)

Generative Pre-trained Transformer (GPT) is an Artificial Intelligence model capable of understanding and generating human-like text based on user input.

Applications:

- *Educational assistance*
- *Question answering*

- *Content generation*
- *Programming support*
- *Report writing*
- *Language learning*

Advantages:

- *Interactive learning*
- *Instant explanations*
- *Personalized assistance*
- *Multiple examples and exercises*

Prompt Engineering

Prompt engineering is the process of creating clear instructions to obtain better responses from AI systems.

Characteristics of a good prompt:

- *Clearly specify the topic*
- *Mention learning level*
- *Specify expected output*
- *Mention examples or questions required*
- *Define response format*

Study Assistant Prompt

A Study Assistant prompt helps students learn topics systematically.

Typical structure:

1. *Subject/topic*
2. *Student level*
3. *Explanation requirement*
4. *Examples*
5. *Practice questions*
6. *Summary*

Sample Study Assistant Prompt

***Prompt**: "Act as a study assistant for diploma Computer Engineering students. Teach the topic Data Structures (Any branch specific topic) in simple language. Explain definitions, concepts, examples, applications, important points, and provide five practice questions with answers."*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab
3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account / Email ID	Valid account for platform login	1 per student
5	GPT Platform	ChatGPT (chatgpt.com) / Google Gemini / Any available GPT tool	1 per student
6	Google Classroom	Accessed via browser	1 per student

PROCEDURE

List the one-by-one procedure to be followed for completing the experiment.

Activity: Creating a Study Assistant Prompt

1. *Open a web browser in Ubuntu.*
2. *Open GPT platform.*
3. *Sign in if required.*
4. *Create a structured prompt.*
5. *Enter prompt into the system.*
6. *Generate response.*
7. *Review the content.*
8. *Refine prompt if necessary.*
9. *Generate additional questions/examples.*
10. *Save observations.*

RESULTS

A structured "Study Assistant" prompt was successfully created and used in a Generative Pre-trained Transformer (GPT) system to learn technical concepts effectively. The GPT-generated explanations, examples, summaries, and practice questions were analyzed

and validated. A report containing the prompt used, generated responses, observations, and necessary screenshots was prepared and successfully uploaded through Google Classroom.

PRACTICAL RELATED QUESTIONS

1. *Expand GPT.*
2. *What is prompt engineering?*
3. *What is a Study Assistant prompt?*
4. *Why should prompts be structured?*
5. *Name one GPT platform.*
6. *What is the advantage of AI-assisted learning?*
7. *Why should AI-generated content be reviewed?*
8. *What is prompt refinement?*
9. *State one application of GPT.*
10. *What is the purpose of follow-up prompts?*

SELF- LEARNING COMPONENTS

List self-learning activities depending on the time allotted in the syllabus.

1. Give ChatGPT this prompt: "Act as a technical interviewer for a Junior Engineer role. Ask me 5 questions one by one about computers."

2. What you mean by C-R-T-O framework in prompt engineering. Generate a proper prompt in the C-R-T-O framework for use cases of your choice.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://www.geeksforgeeks.org/prompt-engineering/	GeeksforGeeks	GeeksforGeeks

Experiment no.: 12	Date:

GRAPHIC DESIGN

PRACTICAL SIGNIFICANCE

To design a technical event poster using AI-assisted design features in Canva Free Version.

RELEVANT PROGRAM OUTCOMES(POS)

PO1: Apply knowledge of mathematics, natural sciences, engineering fundamentals, and an engineering specialization (as specified in DK1 to DK4) to a wide range of practical procedures and practices

PO2: Identify and analyze well-defined engineering problems and reach substantiated conclusions using codified methods of analysis specific to their field of activity (DK1 to DK4).

PO3: Design solutions for well-defined technical problems and assist in the design of systems, components, or processes to meet specified needs, with appropriate consideration for public health and safety, as well as cultural, societal, and environmental factors (DK5).

PO4: Conduct investigations of well-defined problems by locating and search relevant codes and catalogues, and by conducting standard tests and measurements (DK8).

PO7: Understand and commit to professional ethics and the norms of technician practice, including compliance with relevant laws. Demonstrate an understanding of the need for diversity and inclusion (DK9).

RELEVANT COURSE OUTCOMES(COS)

CO4: Apply generative AI, digital design platforms for guided learning and visual communication.

THEORY

Graphic Design

Graphic design is the process of creating visual content to communicate information effectively using text, images, colors, and layouts.

Applications:

- *Event posters*
- *Advertisements*
- *Social media posts*
- *Technical brochures*

- *Presentation graphics*

AI in Graphic Design

Artificial Intelligence tools assist users in creating designs automatically by suggesting:

- *Templates*
- *Images*
- *Layouts*
- *Backgrounds*
- *Design styles*
- *Text content*

Advantages:

- *Reduces design time*
- *Provides professional templates*
- *Improves creativity*
- *Easy customization*

Canva

Canva is an online graphic design platform used for creating visual content such as posters, presentations, banners, and social media graphics.

Features:

- *Drag-and-drop interface*
- *Templates*
- *AI-assisted design tools*
- *Icons and images*
- *Text editing*
- *Export options*

RESOURCES REQUIRED

Sl. No.	Name of resource	Suggested specification	Quantity
1	Computer System	Desktop/Laptop with Ubuntu Linux	1 per student
2	Internet Connection	Broadband / Wi-Fi	1 per lab

3	Web Browser	Mozilla Firefox / Google Chrome	Installed
4	Google Account	Valid institution or personal email ID	1 per student
5	Canva	Free Version, accessed via browser (canva.com)	1 per student
6	College Logo File	PNG format with transparent background (provided by institution)	1
7	Google Classroom	Accessed via browser	1 per student

PROCEDURE

List the one-by-one procedure to be followed for completing the experiment.

Activity: Creating Technical Event Poster

1. *Open Canva Website*

   Open the browser on your computer. Type www.canva.com in the address bar and press Enter.

2. *Sign in.*

   Click "Sign up" if new, or "Log in" if you already have an account. Click "Continue with Google" and select your email ID. You will be taken to the Canva Home Screen.

3. *Select Poster design.*

   Click the search bar at the top, type "Poster" and press Enter. Select "Poster (Portrait)" from the results. Click "Create a blank poster" to open the editor.

4. *Use AI prompt for poster generation.*

   In the left side panel, click "Magic Design" (star/wand icon). A text box will appear. Type a prompt describing your event poster, for example:

"Technical event poster for a symposium on Artificial Intelligence, blue and white color scheme, with space for event name, date, venue, and chief guest details."
Click "Generate". Wait a few seconds. Canva AI will display 4 to 6 ready-made poster designs. Click the design you like best to open it in the editor.
Edit Event Details
Double-click on any text in the poster to edit it. Press Ctrl + A to select the existing text, delete it, and type your own content. Replace all placeholder text with actual event information:
Event name (e.g., TECHNOVA 2025)
Event type (e.g., National Level Technical Symposium)
Date, Time, and Venue
Chief Guest name and designation
Department and College name
Contact email or phone number.

5. *Edit Event Details.*

Double-click on any text in the poster to edit it. Press Ctrl + A to select the existing text, delete it, and type your own content. Replace all placeholder text with actual event information:
Event name (e.g., TECHNOVA 2025)
Event type (e.g., National Level Technical Symposium)
Date, Time, and Venue
Chief Guest name and designation
Department and College name
Contact email or phone number

6. *Format Text*

Click on a text box and use the top toolbar to change the font style, font size, bold/italic, and text color. Use large font size (60–80pt) for the event title and smaller size (18–24pt) for body text. Ensure the text color contrasts clearly with the background.

7. *Add Icons and College Logo*

Click "Elements" in the left panel. Search for keywords like "technology" or "circuit" and click any free icon to add it to the poster. To add your college logo, click "Uploads" → "Upload files", select the logo from your computer, and drag it to the top or bottom of the poster.

8. *Adjust Background*

   Click on the background area of the poster. Use the color box in the top toolbar to change the background color. Choose a color that contrasts well with your text, such as dark blue with white text.

9. *Preview the Poster*

   Click the eye icon (Preview) at the top-right corner. Check for spelling errors, overlapping elements, and missing information. Press Escape to return to the editor and make corrections if needed.

10. *Download the Poster*

    Click "Share" at the top-right, then click "Download". Select PNG for digital submission or PDF for printing. Click "Download" — the file will be saved in your Downloads folder.

11. *Upload to Google Classroom*

    Open Google Classroom, go to the assigned practical, click "Add or Create" → "File", select the downloaded poster file, and click "Turn In".

RESULTS

A technical event poster was successfully designed using AI-assisted design features available in Canva Free Version. The generated design was customized by modifying text, layout, graphics, and visual elements to meet the event requirements. A report containing the design prompt, the final poster was prepared and successfully uploaded through Google Classroom.

PRACTICAL RELATED QUESTIONS

1. *What is graphic design?*
2. *What is Canva?*
3. *What is AI-assisted design?*
4. *What is Magic Design?*
5. *Why are templates used?*
6. *What is the purpose of posters?*
7. *Name two export formats in Canva.*
8. *What is the use of icons in posters?*
9. *Why should color combinations be selected carefully?*
10. *State one advantage of AI in graphic design.*

SELF- LEARNING COMPONENTS

1. Use canva or any web based graphic design platform to design logo for your college art festival

2. Use Canva's Magic Studio or any web based graphic design platform to place a photo of yourself in front of a "High-Tech Server Room" background for a professional poster.

REFERENCES

Sl. No.	Text Book/ Website	Author	Publication
1	https://www.canva.com/en_in/help/use-magic-design/	Canva	Canva Official
2	https://www.geeksforgeeks.org/graphic-design/	GeeksforGeeks	GeeksforGeeks